

Osmania University

SUPPLEMENTARY EXAMINATION ? 2025

Electric Vehicle Technology ? Supplementary Examination, 2025

Subject Code: EE602 Semester: 7 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: EV Architecture)

1. Derive the expression related to EV Architecture and explain each term. [12 Marks]
2. Solve a problem related to EV Architecture. Show all the steps clearly. [12 Marks]
3. Describe the working principle of EV Architecture with necessary equations. [12 Marks]

SECTION B (Unit 2: Battery Systems)

4. Compare and contrast the key concepts involved in Battery Systems. [12 Marks]
5. Describe the working principle of Battery Systems with necessary equations. [12 Marks]
6. How is Battery Systems implemented? Give an algorithm / procedure and explain. [8 Marks]

SECTION C (Unit 3: Electric Motors for EVs)

7. Explain Electric Motors for EVs with a neat diagram/sketch and a suitable example. [12 Marks]
8. List the advantages and limitations of Electric Motors for EVs. Explain any two in detail. [12 Marks]
9. Define Electric Motors for EVs. Explain its significance in Electric Vehicle Technology with a suitable example. [12 Marks]

SECTION D (Unit 4: Charging Infrastructure)

10. Solve a problem related to Charging Infrastructure. Show all the steps clearly. [12 Marks]
11. Derive the expression related to Charging Infrastructure and explain each term. [12 Marks]
12. List the advantages and limitations of Charging Infrastructure. Explain any two in detail. [12 Marks]

SECTION E (Unit 5: Battery Management Systems)

13. List the advantages and limitations of Battery Management Systems. Explain any two in detail. [12 Marks]
14. Solve a problem related to Battery Management Systems. Show all the steps clearly.

[12 Marks]

15. Explain Battery Management Systems with a neat diagram/sketch and a suitable example. [12 Marks]

*** End of Question Paper ***